

作品 1：三層 IP 網路架構 (Routed-Access) 與 Rocky Linux 及 ISO 27001:2022 安全規範整合實作

平台：Cisco IOS/ GNS3 / Rocky Linux

安全框架：27001:2022

關鍵技術：L3 Routed Access, EtherChannel, Syslog, DHCP, HSRP, PAT, NTP, OSPF, Stratis File System (auto mount), ISO 27001:2022 Control Implementation

作品商業背景概述 (第 2 頁)

系統架構與合規性概述 (第 2 頁)

拓樸結構 (第 2 頁)

IP位址表 (第 3 頁)

驗證 PAT 配置 (第 4 頁)

驗證 DHCP 地址釋出 (第 4 頁)

驗證 HSRP 備用路由器 (第 5 頁)

驗證 OSPF-Neighborship with Authentication (第 5 頁)

驗證集中式日誌記錄 (第 6 頁)

驗證 NTP和伺服器時間同步 (第 6 頁)

驗證 Stratis File System (第 7 頁)

遵守 ISO 27001:2022 - A.8.2 - 特權存取權限 & A.8.5 安全認證 (第 7 頁)

遵守 ISO 27001:2022 - A.8.20 - 網路安全 & A.8.9 配置管理 (第 7 頁)

遵守 ISO 27001:2022 - A.5.17 - 身份驗證訊息 & A.8.9 配置管理 (第 7 頁)

遵守 ISO 27001:2022 - A.8.15 - 日誌記錄 (第 8 頁)

遵守 ISO 27001:2022 - A.8.17 - 時鐘同步 (第 8 頁)

遵守 ISO 27001:2022 - A.8.20 網路安全 (第 8 頁)

遵守 ISO 27001:2022 - A.8.24 密碼學的使用 (第 9 頁)

作品商業背景概述：

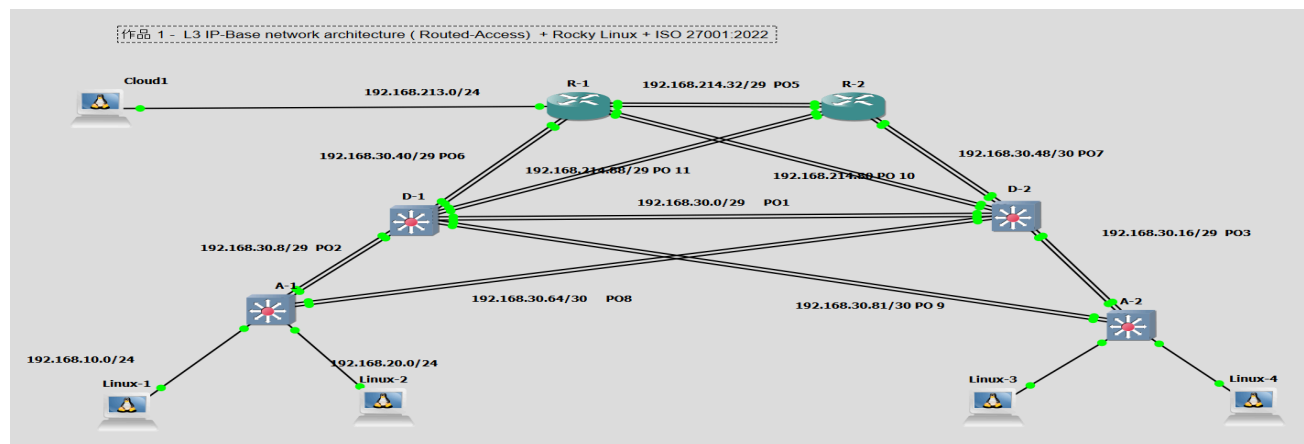
一家區域性金融服務機構正在升級其資料中心基礎設施，以符合ISO 27001:2022 認證要求，同時更換老舊且容易發生故障的二層交換網路。先前的架構存在諸多問題，例如：中斷期間生成樹協定 (STP) 收斂速度慢、頻寬可擴展性有限以及安全控制分散。

為了消除單點故障、滿足審計合規性要求並支援機架的快速擴展，該機構部署了路由存取三層IP架構，並整合了加固型Rocky Linux中央管理主機

系統架構與合規性概述：

- 高可用性與擴充性 (Availability & Scalability)：全網採用全三層路由 (Routed Core & Access) 結合 Port-Channel 多線路冗餘，適合在要求零中斷服務企業環境中。此架構採用 Routed Access (L3 路由到接入層)，透過 OSPF 搭配 ECMP實現毫秒級收斂與流量負載分流，未來新增機架 (Rack) 或樓層，只需分配新的 IP Subnet 並發布路由，不影響原有網路拓撲。配合 Rocky Linux 建立集中式服務 (Syslog 遠端日誌伺服器、NTP 時間同步伺服器、DHCP) 與 Stratis 存取，能有效降低營運維護成本並符合系統 hardening (強化安全) 要求。
- ISO 27001:2022 安全框架控制 (Security Controls)：「ISO 27001:2022 安全控制措施 (Security Controls)：A.5.17 身分驗證資訊 (強制執行使用者密碼生命週期管理與定期輪調、A.8.2 特權存取權 (防止未授權的 Root 權限存取)、A.8.5 安全性認證 (取消 Root 外部直接登入管理，配備 SSHv2 存取權、A.8.15 日誌留存 (Rocky Linux 與 L3 設備集中系統日誌審計)、A.8.17 時脈同步 (確保整體網路架構與伺服器環境系統時脈一致)、A.8.20 網路安全性 (僅保留必要開放端口，並包圍不必要的網路服務與高風險協議) 及 A.8.24 密碼學的使用 (保護網路傳輸與管理界面之資訊機密性)，以確保整體架構符合國際資安標準。

拓撲結構：



IP Address Table

Device	Port	IP	Mask	Description
R1	g1	192.168.213.12	/24	connect to Gateway
	g2	shut down		
	g3	po5		connect to R-2
	g4	po5		connect to R-2
	g5	po6		connect to D-1
	g6	po6		connect to D-1
	g7	po6		connect to D-2
	g8	po6		connect to D-2
	po 5	192.168.30.33	/29	connect to R-2
	po 6	192.168.30.42	/29	connect to D-1
	po 10	192.168.30.81	/29	connect to D-2
	standby 5	192.168.30.35		connect to R-2 (HSRPv2)
	standby 6	192.168.30.46		connect to D-1 (HSRPv2)
	standby 10	192.168.30.86		connect to D-2 (HSRPv2)
D-2	g0/0	shut down		
	g0/1	shut down		
	g0/2	po1		connect to D-1
	g0/3	po1		connect to D-1
	g1/0	po3		connect to A-2
	g1/1	po3		connect to A-2
	g1/2	po7		connect to R-2
	g1/3	po7		connect to R-2
	g2/0	po8		connect to A-1
	g2/1	po8		connect to A-1
	g2/2	po10		connect to R-1
	g2/3	po10		connect to R-1
	g3/0	shut down		
	g3/1	shut down		
	g3/2	shut down		
	g3/3	shut down		
	po 1	192.168.30.2	/29	connect to D-1
	po 3	192.168.30.18	/29	connect to A-2
	po 7	192.168.30.42	/29	connect to R-2
	po 8	192.168.30.52	/29	connect to A-1
	po 10	192.168.30.82	/29	connect to R-1
	standby 1	192.168.30.6		connect to D-1 (HSRPv2)
	standby 3	192.168.30.22		connect to A-2 (HSRPv2)
	standby 7	192.168.30.46		connect to R-2 (HSRPv2)
	standby 8	192.168.30.52		connect to A-1 (HSRPv2)
	standby 10	192.168.30.86		connect to R-1 (HSRPv2)

驗證 PAT 配置

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp  192.168.213.12:5068 192.168.10.100:33732 34.49.197.38:443   34.49.197.38:443
tcp  192.168.213.12:5067 192.168.10.100:33722 34.49.197.38:443   34.49.197.38:443
tcp  192.168.213.12:5062 192.168.20.110:56328 34.49.197.38:443   34.49.197.38:443
tcp  192.168.213.12:5069 192.168.10.100:33742 34.49.197.38:443   34.49.197.38:443
Total number of translations: 4
```

驗證 DHCP 地址釋出 Linux-1 and Linux-2

- From Rocky Linux (DHCP server - IP release to Linux-1 & Linux-2)

```
Aug  8 10:39:42 localhost kea-dhcp4[4967]: 2026-08-08 10:39:42.696 INFO [kea-dhcp4.leases
/4967.140352074102464] DHCP4_LEASE_REUSE [hwtype=1 0c:ca:02:c8:00:00], cid=[01:0c:ca:02:c8
:00:00], tid=0x19684f9e: lease 192.168.20.110 has been reused for 6727 seconds
Aug  8 10:49:02 localhost kea-dhcp4[4967]: 2026-08-08 10:49:02.031 INFO [kea-dhcp4.leases
/4967.140352057317056] DHCP4_LEASE_ALLOC [hwtype=1 0c:e9:b4:e4:00:00], cid=[01:0c:e9:b4:e4
:00:00], tid=0x6926f326: lease 192.168.10.100 has been allocated for 7200 seconds
```

```
[root@rocky-cloud ~]# nmcli dev show | grep IP4
IP4.ADDRESS[1]: 192.168.10.100/24
IP4.GATEWAY: 192.168.10.1
IP4.ROUTE[1]: Linux-1 obtain DHCP IP from Server dst = 192.168.10.0/24
IP4.ROUTE[2]: dst = 0.0.0.0/0, nh
IP4.DNS[1]: 8.8.8.8
IP4.ADDRESS[1]: 127.0.0.1/8
IP4.GATEWAY: --
[root@rocky-cloud ~]#
```

```
[root@rocky-cloud ~]# nmcli dev show | grep IP4
IP4.ADDRESS[1]: 192.168.20.110/24
IP4.GATEWAY: Linux-2 obtain 192.168.20.1
IP4.ROUTE[1]: DHCP IP from dst = 192.168.20.0/24, nh = 0.0.0.0, mt = 100
IP4.ROUTE[2]: Server dst = 0.0.0.0/0, nh = 192.168.20.1, mt = 100
IP4.DNS[1]: 8.8.8.8
IP4.ADDRESS[1]: 127.0.0.1/8
```

驗證 HSRP 備用路由器

```
D-2#sh standby br
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Po1	1	100		Standby	192.168.30.1	local	192.168.30.6
Po3	3	110	P	Active	local	192.168.30.18	192.168.30.22
Po7	7	100		Standby	192.168.30.41	local	192.168.30.46
Po8	8	110	P	Active	local	192.168.30.50	192.168.30.52
Po10	10	100		Standby	192.168.30.81	local	192.168.30.86

P indicates configured to preempt.
|

```
.
interface Port-channel3
 no switchport
 ip address 192.168.30.19 255.255.255.248
 standby version 2
 standby 3 ip 192.168.30.22
 standby 3 priority 110
 standby 3 preempt
 ip ospf authentication key-chain OSPF-KEY
 ip ospf 100 area 0
```

驗證 OSPF-Neighborship with Authentication

```
D-2#sh ip os neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.213.12	1	FULL/DR	00:00:38	192.168.30.81	Port-channel10
192.168.30.89	1	FULL/DR	00:00:38	192.168.30.41	Port-channel7
192.168.30.75	1	FULL/BDR	00:00:35	192.168.30.18	Port-channel3
192.168.30.90	1	FULL/DR	00:00:38	192.168.30.1	Port-channel1

D-2#

```
key chain OSPF-KEY
 key 200
   key-string 7 0110050A5A
   cryptographic-algorithm hmac-sha-256
 password encryption aes
```

驗證集中式日誌記錄 (Centralize Log Server)

```
root@localhost:/var/log/remote# ls
127.0.0.1      192.168.30.1  192.168.30.18  192.168.30.75
192.168.213.12 192.168.30.11 192.168.30.19
root@localhost:/var/log/remote# cd 192.168.30.1
root@localhost:/var/log/remote/192.168.30.1# ls
2026-08-08.log 2026-08-09.log
root@localhost:/var/log/remote/192.168.30.1# cat 2026-08-08.log
Aug  8 20:12:37 192.168.30.1 139: *Aug  8 10:05:36.050: %OSPF-5-ADJCHG: Process 100, Nbr 1
92.168.215.1 on Port-channel11 from LOADING to FULL, Loading Done
root@localhost:/var/log/remote/192.168.30.1# cd ..
root@localhost:/var/log/remote# cd 192.168.30.19
root@localhost:/var/log/remote/192.168.30.19# ls
2026-08-08.log
root@localhost:/var/log/remote/192.168.30.19# cat 2026-08-08.log
Aug  8 18:46:29 192.168.30.19 103: *Aug  8 09:56:00.424: %GRUB-5-CONFIG_WRITING: GRUB conf
iguration is being updated on disk. Please wait...
Aug  8 18:46:29 192.168.30.19 104: *Aug  8 09:56:01.121: %GRUB-5-CONFIG_WRITTEN: GRUB conf
iguration was written to disk successfully.
```

```
A-2(config)#logg
A-2(config)#logging on
A-2(config)#logging host 192.168.213.14
A-2(config)#logging h
*Aug  8 06:07:13.164: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.213.14 port 514 started - CLI initiate
% Incomplete command.

A-2(config)#logging tr
A-2(config)#logging trap 5
```

驗證 NTP和伺服器時間同步

```
#
# add network in /etc/chrony.conf
allow 192.168.213.0/24
allow 192.168.30.0/24
```

```
root@localhost:~# chronyc sources -v
```

```

.-- Source mode  '^' = server, '=' = peer, '#' = local clock.
/ .-- Source state '*' = current best, '+' = combined, '-' = not combined,
| /           'x' = may be in error, '~' = too variable, '?' = unusable.
||
||           Reachability register (octal) --.          | xxxx [ yyyy ] +/- zzzz
||           Log2(Polling interval) --.          | |      | xxxx = adjusted offset,
||                                     \   |      | |      | yyyy = measured offset,
||                                     |   |      | |      | zzzz = estimated error.
||                                     |   |      | \
||                                     |   |      |
MS Name/IP address             Stratum Poll Reach LastRx Last sample
=====
^+ 103.186.118.217                2    6    17    32    +593us[ +593us] +/- 8188us
^- 123-204-232-128.adsl.dyn>     2    6    17    32   +1365us[+1365us] +/-   56ms
^- 103.186.118.222                2    6    17    32    -475us[ -475us] +/-   13ms
^* 103.186.118.212                2    6    17    32    -172us[-1553us] +/- 9388us
```

```
D-1#sh ntp associations
```

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.213.14 103.186.118.212 3    22    64    3  7.205 4761.23 1937.8
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

驗證Stratis File System : auto mount to client

```
root@localhost:~# stratis filesystem list
```

Pool	Filesystem	Total / Used / Free / Limit	Device	UUID
pool1	fs1	1 TiB / 546 MiB / 1023.47 GiB / None	/dev/stratis/pool1/fs1	3dac3735-2032-4702-b249-32cca0293a46
pool1	snapshot1	1 TiB / 546 MiB / 1023.47 GiB / None	/dev/stratis/pool1/snapshot1	41202873-6a39-474a-802a-ed24550098dd

```
[root@rocky-cloud ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/vda4	8.4G	1.1G	7.4G	13%	/
devtmpfs	594M	0	594M	0%	/dev
tmpfs	633M	0	633M	0%	/dev/shm
tmpfs	254M	552K	253M	1%	/run
tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
/dev/vda3	936M	279M	658M	30%	/boot
/dev/vda2	200M	14M	186M	7%	/boot/efi
tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
tmpfs	127M	4.0K	127M	1%	/run/user/1000
192.168.213.14:/srv/stratis_share	17G	5.4G	12G	32%	/mnt/client_stratis

遵守 ISO 27001:2022 - A.8.2—特權存取權限 (要求對特權存取權限的分配和使用進行嚴格控制和限制), A.8.5—安全認證 (緩解針對管理員帳戶的暴力破解攻擊)

```
root@localhost:~# grep -i "PermitRootLogin" /etc/ssh/sshd_config
#PermitRootLogin prohibit-password
# the setting of "PermitRootLogin prohibit-password".
PermitRootLogin no
```

遵守 ISO 27001:2022 : A.8.20—網路安全 (最少的開放網路端口), A.8.9— 配置管理 (服務最小化)

```
root@localhost:~# firewall-cmd --list-services
cockpit dhcp dhcpv6-client mountd nfs ntp rpc-bind ssh
```

遵守 ISO 27001:2022 : A.5.17—身份驗證訊息 (強制執行使用者密碼生命週期管理和定期輪調), A.8.9 配置管理(強制執行標準化的 Linux 帳戶老化和密碼原則基準)

```
root@localhost:~# chage --maxdays 60 peter
root@localhost:~# chage --list peter
```

Last password change	: Aug 09, 2026
Password expires	: Oct 08, 2026
Password inactive	: never
Account expires	: never
Minimum number of days between password change	: 0
Maximum number of days between password change	: 60
Number of days of warning before password expires	: 7

遵守 ISO 27001:2022 : A.8.15—日誌記錄 (產生、匯總和儲存遠端事件日誌，以便進行分析和稽核)

```
root@localhost:/var/log/remote# ls
127.0.0.1 192.168.213.12 192.168.30.1 192.168.30.11 192.168.30.18 192.168.30.19 192.168.30.75
root@localhost:/var/log/remote# cat 192.168.30.11
cat: 192.168.30.11: Is a directory
root@localhost:/var/log/remote# cd 192.168.30.11
root@localhost:/var/log/remote/192.168.30.11# ls
2026-08-08.log 2026-08-09.log
root@localhost:/var/log/remote/192.168.30.11# cat 2026-08-09.log
Aug 9 14:11:39 192.168.30.11 93: *Aug 9 06:11:25.889: %HSRP-5-STATECHANGE: Port-channel8 Grp 8 state Sp
eak -> Standby
Aug 9 14:12:01 192.168.30.11 94: *Aug 9 06:11:47.447: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vi
os_l2-adventerprisek9-m' passed code signing verification
Aug 9 15:41:42 192.168.30.11 98: *Aug 9 07:40:17.045: %SYS-5-CONFIG_I: Configured from console by conso
le
Aug 9 15:44:25 192.168.30.11 101: *Aug 9 07:43:00.584: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was w
ritten to disk successfully.
root@localhost:/var/log/remote/192.168.30.11# S
```

遵守 ISO 27001:2022 : A.8.17—時鐘同步 (確保整個環境中的系統時鐘與標準時間源保持一致)

```
R-2#sh ntp associations
```

```
address      ref clock      st  when  poll reach  delay  offset  disp
*~192.168.213.14 103.186.118.215 3    259   256   377  1.799 1116404 3.329
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

```
D-1#sh ntp associations
```

```
address      ref clock      st  when  poll reach  delay  offset  disp
*~192.168.213.14 103.186.118.212 3    22    64    3   7.205 4761.23 1937.8
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

遵守 ISO 27001:2022 : A.8.20 網路安全 (關閉阻止不需要的網路服務)

```
no cdp run
no ip http server
no ip http secure-server
```

```
no ip source-route
```

遵守 27001:2022 - A.8.24 密碼學的使用 (有效地使用加密技術進行控制，以保護資訊的機密性、真實性和完整性。)

```
no ipv6 cef
!
key chain OSPF-KEY
  key 200
    key-string 7 110A1A0B16
    cryptographic-algorithm hmac-sha-256
password encryption aes
```

```

R-2#sh run | s line
line con 0
  stopbits 1
line vty 0 4
  password 7 022527753B
  login
  transport input ssh
line vty 5 15
  password 7 022527753B
  login
  transport input ssh
```